

COURSE TITLE : INDUSTRIAL SAFETY
COURSE CODE : 5073
COURSE CATEGORY : E
PERIODS/ WEEK : 4
PERIODS/ SEMESTER : 52
CREDIT : 4

TIME SCHEDULE

MODULE	TOPIC	PERIOD
1	understand the safety in process industries.	13
2	Know the personal protective equipments and Hazardous properties of chemicals and metals.	13
3	Appreciate fire detection, prevention, fire fighting in industries.	13
4	Understand safety management.	13
TOTAL		52

COURSE OUTCOME:

SL.NO	SUB	STUDENT WILL BE ABLE TO
1	1	understand the safety in process industries.
2	2	Know the personal protective equipments and Hazardous properties of chemicals and metals.
	3	Appreciate fire detection, prevention, fire fighting in industries.
	4	Understand safety management.

SPECIFIC OUTCOME

MODULE I

1.1.0 Understand safety in chemical process

- 1.1.1 Define hazard and explain the classifications of hazards
- 1.1.2 Explain and classify potentially dangerous processes with examples.
- 1.1.3 Explain the Importance of vessel design and basic information needed for vessel design.
- 1.1.4 Describe Causes and prevention of pressure vessel hazards.
- 1.1.5 Define explosive hazard,
- 1.1.6 State Hazard Code and Explosive Limit,
- 1.1.7 Differentiate Lower Explosive Limit and Upper Explosive Limit.
- 1.1.8 Define Toxic Hazard and Toxicity,
- 1.1.9 List Threshold Limit Values of toxic chemicals.
- 1.1.10 Define Flammability
- 1.1.11 Classify flammable liquids in terms of flash point.

- 1.1.12 State 'S' classification of chemicals
- 1.1.13 Describe Threshold Limit Value(TLV), Time Weighted Average(TWA), Maximum Allowable Concentration(MAC), Short Term Explosive Limit(STEL), Frequency Rate(FR), Severity Rate(SR),Frequency Severity Rate (FSR).
- 1.1.14 Describe safety measures to be taken in handling Gases, Liquids and Solids.
- 1.1.15 Explain safety handling of Chlorine Gas, Ammonia and LNG.
- 1.1.16 Explain storage of flammable Solids ,Liquids and Hydrocarbons and
- 1.1.17 Describe the procedures in Transporting Hazardous chemicals.

MODULE II

- 2.1.0 **Know PERSONAL PROTECTIVE EQUIPMENTS AND HAZARDOUS PROPERTIES OF CHEMICALS AND METALS**
 - 2.1.1 EXPLAIN Requirements of personal protective Equipment.
 - 2.1.2 Explain Classification and Selectivity of personal protective Equipment.
 - 2.1.3 Explain Personal protective Equipment for sound and vibration.
 - 2.1.4 Explain Personal protective Equipment for power plants.
 - 2.1.5 State Environmental Hazards.
 - 2.1.6 State Radiation Hazards.
 - 2.1.7 State Mechanical Hazards.
 - 2.1.8 Explain Machine guards and safety devices.
 - 2.1.9 Explain Static electricity hazards.
 - 2.1.10 Explain Pressure vessel hazards.
 - 2.1.11 List Permissible limits of chemicals in water and air.
 - 2.1.12 Explain Relief systems in process plants.
 - 2.1.13 Explain Hazard assessment.
 - 2.1.14 List about the hazardous properties of chemicals like Acetonitrile, Butylamine, Benzene, Acetaldehyde, Monochloro acetic acid, Butane, Caprolactum, Acetone, Cyclohexanol, Hexane, Phenol and Toluene.
 - 2.1.15 Explain hazards of metals like Arsenic, Cadmium, Barium, Copper, Mercury, Selenium, Tin and Zinc.

MODULE III

- 3.1.0 **COMPREHEND FIRE DETECTION, PREVENTION, FIRE FIGHTING IN PLANTS**
 - 3.1.1 Describe the chemistry of fire,
 - 3.1.2 Explain elements of fire and fire triangle.
 - 3.1.3 Describe slow combustion, rapid combustion & spontaneous combustion
 - 3.1.4 Classify fire according to combustible materials
 - 3.1.5 List the causes of initiation of fire.
 - 3.1.6 List the causes of electrical fire.

3.2.0 Understand fire detectors

- 3.2.1 Explain the Criteria for selection of fire detectors.
- 3.2.2 Explain about the following types of detectors in chemical plants
smoke detector, ionisation detector, photoelectric detector, infrared detector.
- 3.2.3 Explain in detail about Fire Alarms.
- 3.2.4 Explain Fire extinguishing technique.
- 3.2.5 List the Indian standards on fire extinguishers.
- 3.2.6 Explain the working of the following extinguishers.
 - 1. Soda Acid extinguishers
 - 2. Gas cartridge and stored pressure water filled extinguisher
 - 3. Antifreeze extinguishers
 - 4. Pump tank extinguishers
 - 5. Foam extinguishers
 - 6. Co₂ extinguishers
 - 7. Dry chemical fire extinguishers
- 3.2.7 List precautions for prevention of fire and describe active fire protection systems.

MODULE IV

4.1.0 UNDERSTAND SAFETY MANAGEMENT SYSTEM

- 4.1.1 Explain about safety training and the role of training in industry.
- 4.1.2 Explain the role of human error in industry.
- 4.1.3 State the general safety rules and describe safety measures in chemical laboratories and Pilot Plants.
- 4.1.4 Explain Hazard check list and Explain safety check list for identification of Hazards
- 4.1.5 Explain safety check list during installation, during start up and during shut down.
- 4.1.6 Explain personal safety and the principles of safety management.
- 4.1.7 Explain Employees state Insurance Act.
- 4.1.8 Explain Workmen's Compensation Act.
- 4.1.9 Understand the following case studies.
 - 1. Benzene leakage from a reboiler.
 - 2. Accident with Trichloroethylene.
 - 3. Hazard caused due to phenol.
 - 4. Accident due to runaway reaction in reactor.
 - 5. Flixborough disaster.
 - 6. Bhopal gas disaster.

CONTENT DETAILS

MODULE I

Hazard-classification –hazardous properties of chemical-flammability upper explosive limit-lower explosive limit- toxicity –hazard codes-concentration limit. Safety in handling gases, Liquids and Solids

MODULE II

Personal protective equipments for sound and vibration- power plants-environmental Hazards. Radiation Hazards-Mechanical Hazards-Machine guards and safety devices-Static electricity hazards-Pressure vessel hazards-Permissible limits of chemicals in water and air-Relief systems in process plants-Hazard assessment.

MODULE III

Chemistry of fire-classification of fire-fire extinguishing techniques-classification-portable fire extinguishers-Indian standard for extinguishers- fire fighting extinguishers

MODULE IV

Safety management-safety training-hazardous check list-safety check list for identification of hazard-safety check list for start up-shut down-installation. Some important case studies.

REFERENCE

- | | | |
|-------------------------|---|--|
| Dr. Sreekant D. Dawande | - | Chemical Hazards and Safety |
| R K Jain, Sunil S. Rao | - | Industrial Safety in Health and Environmental Management Systems |